

OPERATOR'S MANUAL

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CHAPTER SIX MAINTENANCE

Introduction

This chapter contains preventive maintenance procedures and instructions for how to replace the flash lamp, set the system timing and change the filters in the water cooling system. The chapter concludes with a listing of instruments which will aid in the maintenance the Infinity laser system.

It is advisable that the optics in the laser are checked for burn marks, dust and scratches once a month. By the time a problem with an optical component is severe enough to affect the laser's performance it is usually too late to restore it.

Optics Cleaning and Inspection

The most common reason for performance degradation in laser systems is contaminated optics. Laser optical components are routinely exposed to very high energy levels. When optical coatings are clean, this energy will either be reflected or transmitted. If the surface is contaminated, part of the energy will be absorbed, creating hot-spots which can burn the precision coating and dramatically reduce reflection efficiency. Absorption caused by contaminated optical surfaces will degrade performance and shorten component life. Once absorption starts to take place, the rate of surface destruction rapidly increases. Thus even a very small contamination can rapidly give rise to considerable damage. Contaminants which can cause absorption include a variety of particles which may fall on the optical surface or condense from surrounding vapors. Even lens tissue fibers, and plastic gloves can be a source of contaminants. In order to maintain optimum performance, proper cleaning is an absolute necessity. Even though it is advisable to regularly clean the optics, there is also a risk of scratching the components in the process. Operating the laser with the cover on will reduce the need to clean the optics. It is also a good habit to keep the laboratory as clean as possible.

Optics Inspection

A small flashlight is useful for optics inspection. Illuminate the surfaces at various angles and look for streaks and dust particles. If the surface does not appear uniformly colored or if there are signs of dust on the surface, use the cleaning procedure for installed optics outlined below.

General Cleaning Procedure

For Installed Optics:

1. Remove the laser head cover.
2. Neatly fold a sheet of lens tissue into a rectangular shape (approximately 1/4 in x 3/4 in.). Take care not to contaminate the tissue with soiled or oily fingers.

3. Clamp and hold the tissue with hemostats. To avoid damage to the optics, ensure that the hemostats are not close to the edge of the tissue.
4. Moisten the tissue with a few drops of reagent grade methanol or acetone.
5. Gently wipe across the optic in one direction, see Figure 6-1. One pass should be sufficient. Repeat if necessary with a clean tissue.

This procedure should work well for most components in the laser. Different procedures are required for optical components that are difficult to access.

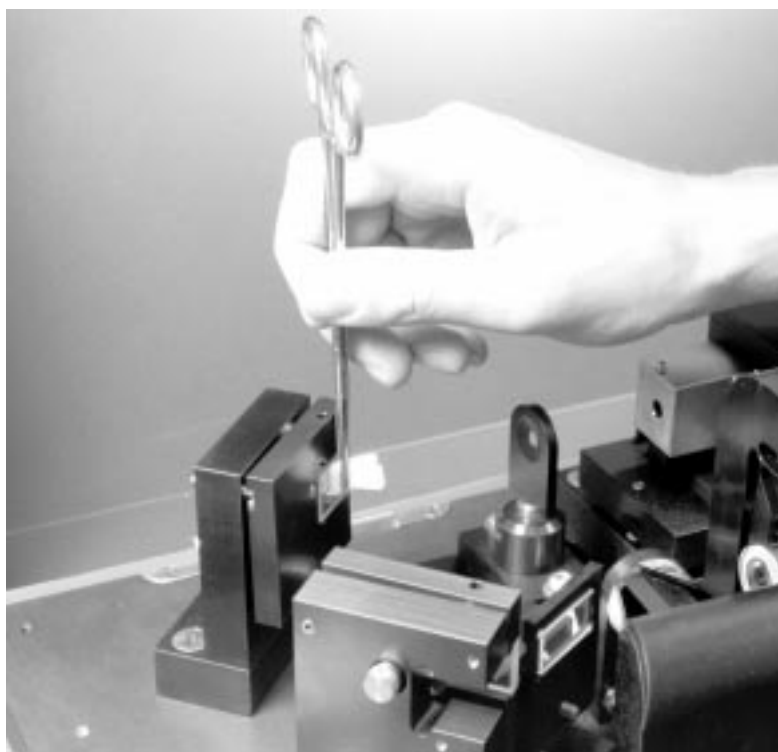


Figure 6-1. Cleaning of Installed optics

90° Polarization Rotator

The plate holding the 90° rotator can be taken out of the laser by removing the spring loaded tensioning screw (see Chapter Three) that holds the moveable part of the mount. The optic can thereafter be cleaned using the procedure described above, see Figure 6-2. Monitor the amount of depolarized energy in the Monitors window after the plate has been reinserted. Adjust the vertical and horizontal tilt angle controls of the mount to bring down the depolarized energy if necessary.



Figure 6-2. Cleaning of 90° Rotator

Relay Imaging Lenses

Removing the vacuum cell in the relay imaging system provides access to the 1 inch concave lens and the relay lens surfaces facing it, Figure 6-3. The windows of the vacuum cell can also be cleaned while the cell is removed. Monitor the amount of depolarized energy in the “Monitors” window after the cell has been reinserted. Follow procedure 4 in Chapter Four to minimize the depolarized energy if it is above 1.5 % of the IR output power.

When reinstalling the vacuum cell, the teflon set screw should not be tightened with a ball driver or Allen key (i.e., the set screw should be “finger” tight). Over tightening the set screw could easily cause the cell to crack.

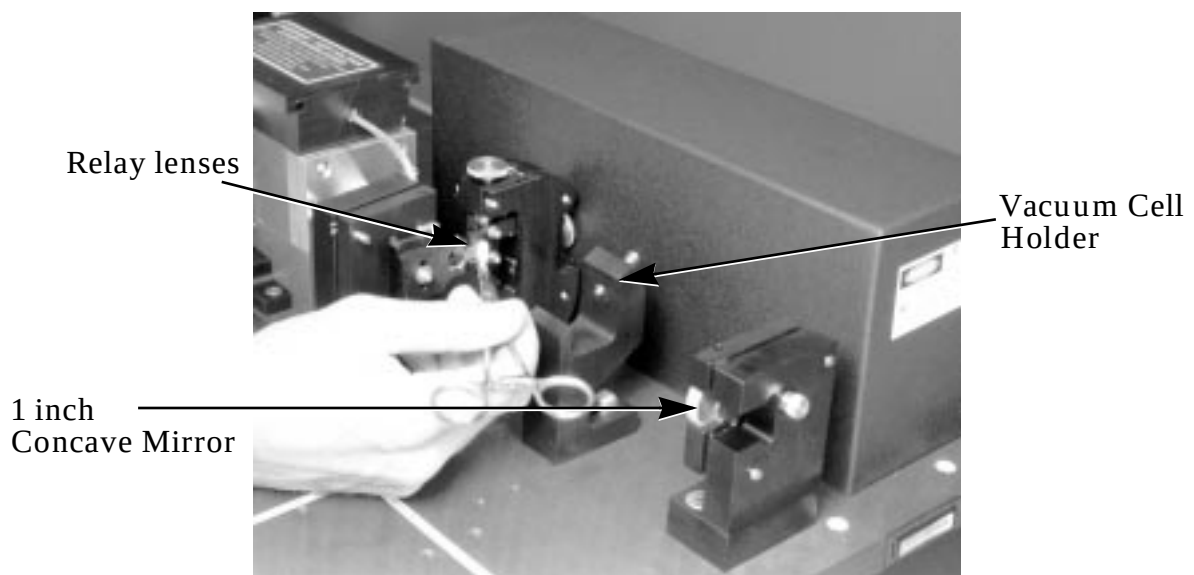


Figure 6-3. Cleaning of Relay Lens #1

Water Cooling System

The water cooling system removes excess heat from the flash lamp, the laser rods and from some of the electrical components in the power supply and the laser head. If the cooling system were to malfunction, the temperature of the laser system would increase, potentially causing irreversible damage. In order to protect the laser from such an event, several safety features have been built into the system. Sensors located at different places in the water loop continuously monitor water level, temperature, flow and resistivity (see Chapter Three, Controls and Indicators). If any of these readings fall outside their preset safety ranges, the microprocessor in the power supply will turn the system off.

To ensure optimal performance and prevent unnecessary downtime it is advisable to periodically check water hoses and water connections for wear and leaks. Make sure that there is enough water in the internal cooling loop. The main reservoir should be at least 90% full. If necessary refill it with “steam” distilled water.

The capacity of the deionization cartridge decreases with time. It should be replaced when it is no longer possible to keep the water resistivity within the desired range, 2 – 6 MΩ cm, by adjusting the flow through the cartridge. See Chapter Three for information about how to adjust the water resistivity. It is advisable to replace the particle filter and the water when the deionization cartridge is replaced.



Use only steam distilled water in the internal water cooling system. Do not use deionized tap water.

For optimum system performance, it recommended that the water and deionization filters be replaced yearly.

Changing the Water and Deionization Filter

1. Turn the key switch on the power supply to off and turn off the main facility circuit breaker. Unscrew and remove the larger of the two covers on the power supply.

Particle Filter Replacement:

- a. Unscrew the filter holder, item 1 in Figure 3-7, and remove the filter inside.
- b. Rinse the new filter thoroughly in distilled water
- c. Insert the new filter and screw the holder back on. It helps if the O-ring is moistened with water first.

Deionization Cartridge Replacement:

- a. Remove the water hoses from the top and bottom of the old cartridge, item 7 in Figure 3-7.
- b. Insert a new cartridge, flow arrow pointing up, and connect the water hoses.
2. Turn on the water pump in the Service window, accessed by clicking on “Service...” in the Infinity control window. Check for leaks. Let the water run until all sounds of bubbles in the system have disappeared and there are no bubbles in the water flowing through the particle filter.
3. Allow the system to run for approximately 1 hour. During this time adjust the deionization cartridge bypass valve, item 3 in Figure 3-7, as necessary to regulate the water resistivity between 2 – 6 MΩ cm.
4. Reinstall the heat exchanger cover. The laser is now ready for normal use.



If the internal water cooling system is drained, remove the deionization cartridge and seal it so that it does not dry out. If the cartridge is allowed to dry it will lose its capacity to maintain a high enough water resistivity. Do not insert a new cartridge until you are ready to fill the system with distilled water.

Change the flash lamp when specified power can no longer be achieved or if the laser energy exhibits large shot to shot fluctuations. Before changing the flash lamp, verify that the shot counter reading is in agreement with the expected life time for the lamp, approximately 50 million shots. If this is not the case there might be a different reason for the performance degradation. See Chapter Four, Procedures 2 and 3, for more information.

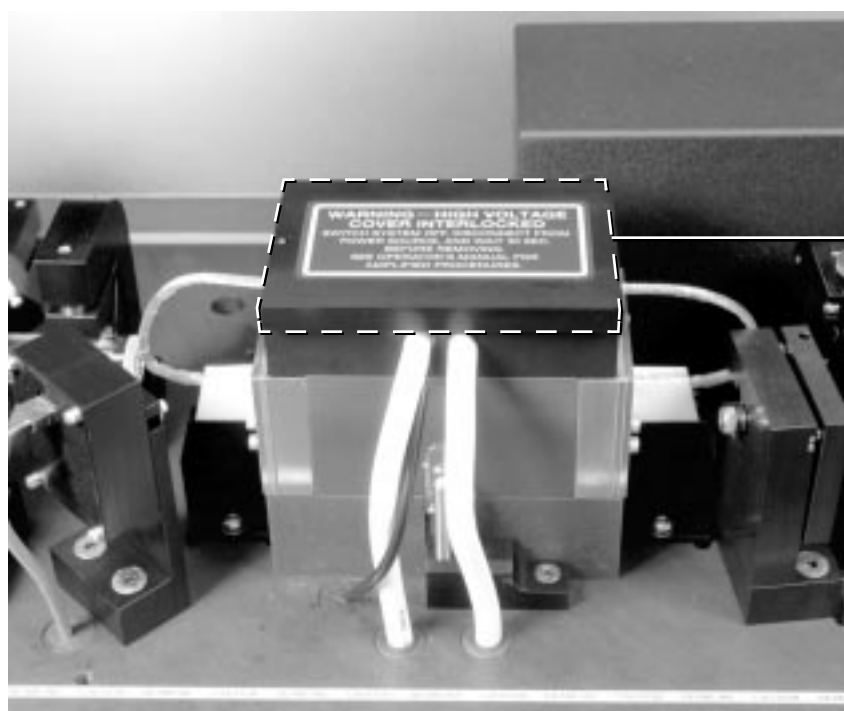


When changing the flash lamp avoid skin to glass contact as this may leave grease marks! Contaminating the Nd:YAG rod end surfaces with finger grease will lead to a drastic increase in absorption that might damage the coatings. Dirt on the glass envelope will reduce the flash lamp efficiency.

Removing the Old Flash Lamp.

Please complete, and check off, each step before proceeding. Numbers in the text are given with reference to Figure 6-4.

1. Turn the power supply key switch to Off position and switch off the main facility circuit breaker.
2. Expose the lamp terminals by sliding the top plastic cover to the side, Figure 6-4.



Slide top cover off to expose flash lamp leads.

Make sure that the power supply is turned off and disconnected from the main AC line!

Figure 6-4. The Infinity POT Assembly with the Flash Lamp Lead Housing on Top

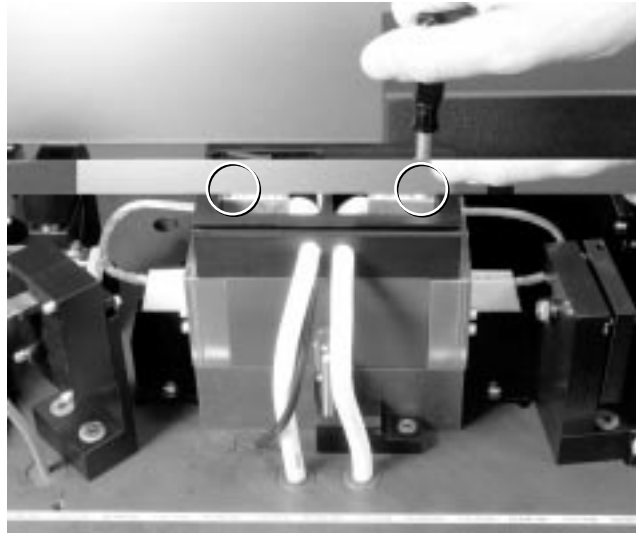


Figure 6-5. Loosening the Flash Lamp Terminal

3. Loosen the lamp terminal, encircled in Figure 6-5, on both sides using the 3/16" nut driver from the maintenance kit.
4. Remove leads from side holes and bend the lamp terminal box on top of the POT out of the way, Figure 6-6.

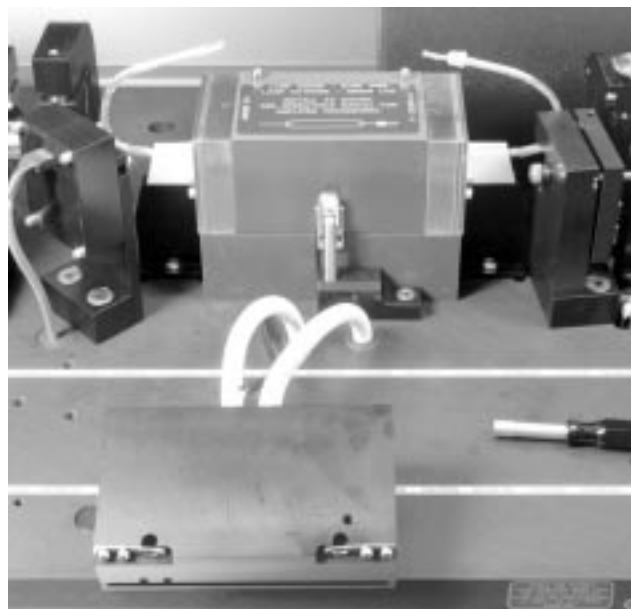


Figure 6-6. Flash Lamp Leads Released and the Lamp Terminal Housing Bent to the Side

5. Prepare a work area with a soft surface on which the POT can be positioned.

6. Remove the two vertical dowel pins, 1 in Figure 6-7, release the latches holding the POT, 2 in Figure 6-7, tilt it sideways a few degrees to drain the water out of it and then lift the POT out of the laser, 3 in Figure 6-7.
7. Put the POT on a clean soft surface so that the green lamp lug (cathode) is on the right-hand side.



Be careful not to scratch the bottom surface of the POT! Doing so might impair the proper function of the two O-ring seals.

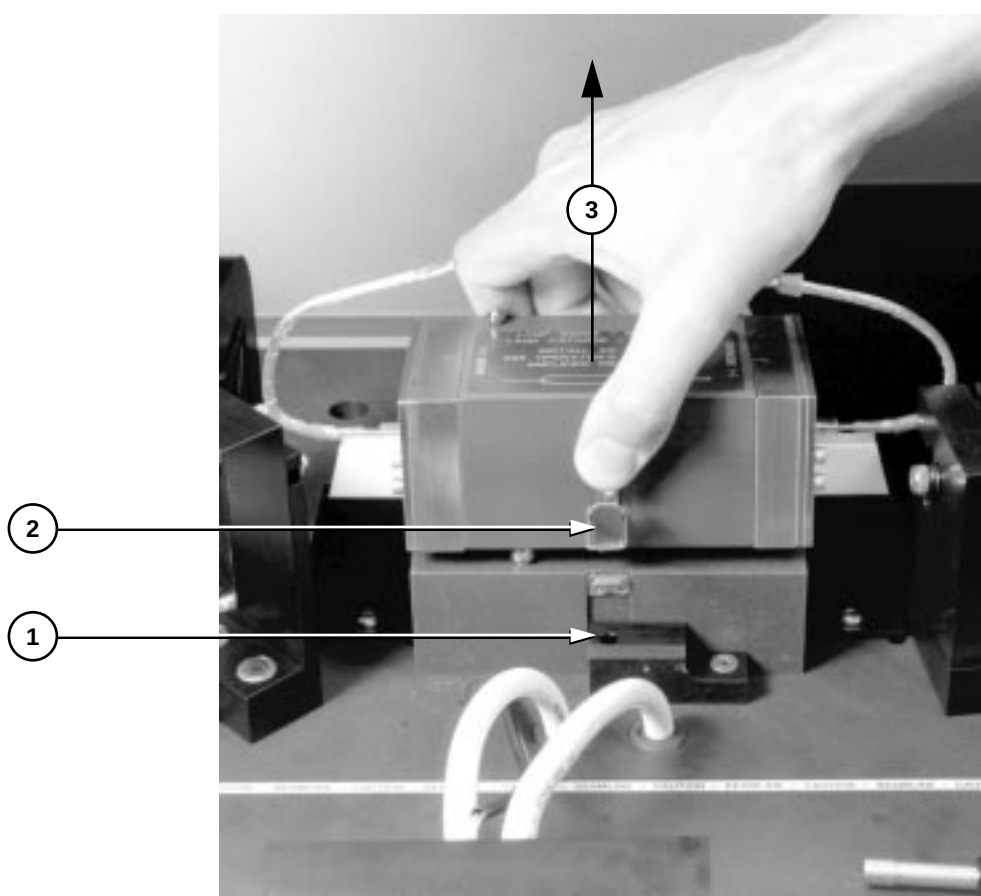


Figure 6-7. Dowel Pins Removed and Latches Released.

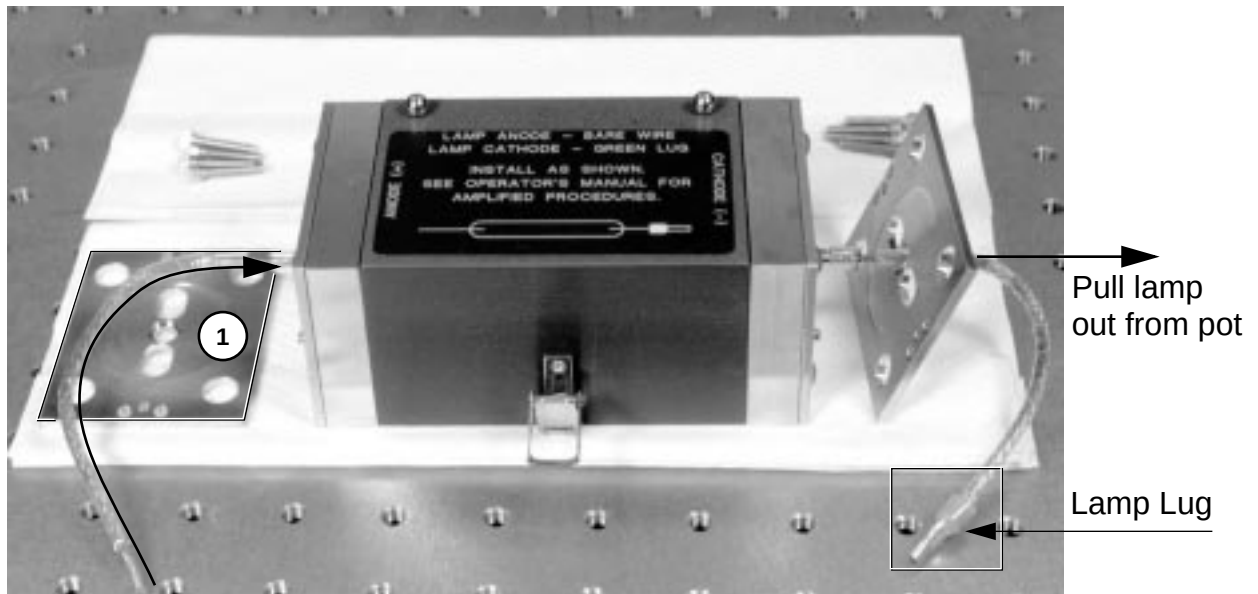


Figure 6-8. POT with the Left Endplate Removed

8. Remove the left endplate, 1 in Figure 6-8. Note that because of the lug attached to the flash lamp lead on the right side the lamp must be taken out of the pot before the right endplate can be removed, 2 in Figure 6-9.
9. Using the finger cots supplied, carefully remove the O-ring on the left-hand side.
10. Slide the flash lamp out of the POT to the right, Figure 6-8. Slide off the remaining O-ring and thereafter the second endplate, 1 and 2 in Figure 6-9, respectively.

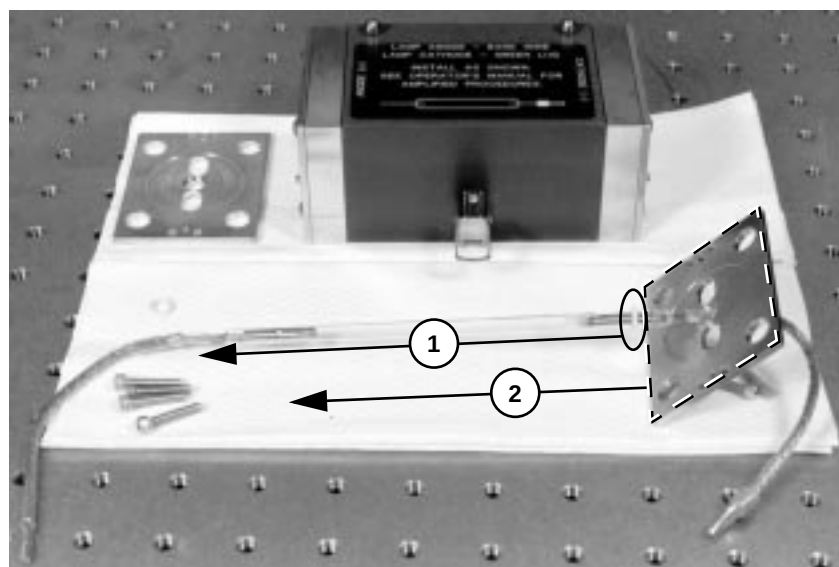


Figure 6-9. Remove the O-ring and the Endplate as Shown

Inserting the New Flash Lamp

Open the box with the replacement lamp and check that the following items are present: 1 Flash lamp, 2 O-rings, 1 O-ring seater and 6 Finger cots.

Please complete, and check off, each step before proceeding.

1. Put on the finger cots supplied with the flash lamp.
2. Take the flash lamp out of the plastic bag.
3. Thread the lug-free end through the right end plate. The side on which the lamp hole is beveled goes against the POT, Figure 6-10.
4. Thread one O-ring over the left lead and slide it across the lamp to a position about 5 mm from the right end of the glass envelope, Figure 6-10. Use the piece of plastic tubing supplied with the flash lamp to push the O-ring across the lamp envelope.
5. Slide the lamp through the POT.
6. Align the plate and POT side slots. Screw the right endplate back on, finger tight.
7. Use the plastic tubing to push the other O-ring into place on the left side, Figure 6-11.
8. Put the left endplate back in place, with the slot on the side of it aligned with that on side of the POT.
9. Center the lamp in the POT such that an equal amount protrudes on each side.

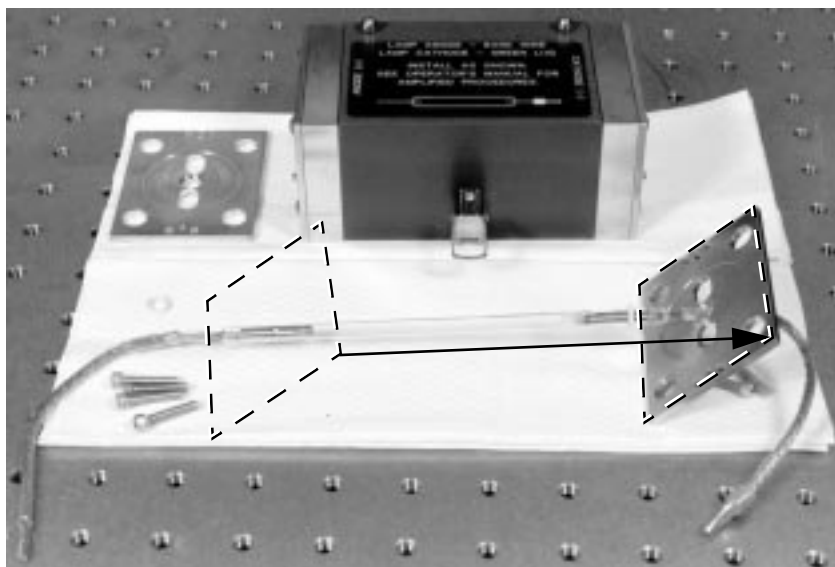


Figure 6-10. Position the Endplate and the O-ring as Shown

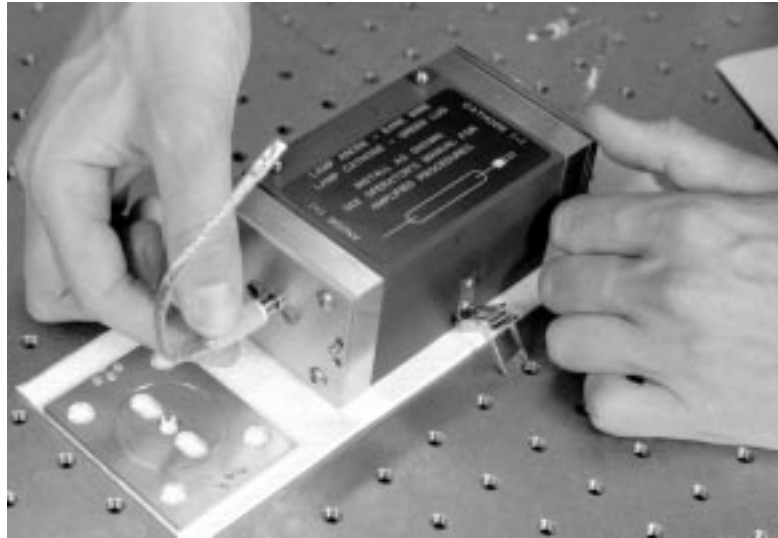


Figure 6-11. Position the Second O-ring as Shown Using the Plastic Tubing Supplied with the Flash Lamp

10. Tighten the screws crosswise, on both sides.
11. Put the POT back into the laser head. Close the latches and make sure that they are properly secured before inserting the dowel pins.
12. Push the wires into the terminal box as far as possible and tighten the 3/16" nuts firmly.
13. Put the lamp terminal box cover back on.
14. Turn on the water pump and check for leaks before starting the laser.
15. Increase the power slowly and, if necessary, minimize the depolarized energy. See Chapter Four, procedure 4, for additional information.

Replacing the EPROM

The Infinity software comprises two separate programs:

1. The Control Program, which is stored permanently on an EPROM inside the power supply and
2. The Graphical User Interface program, INFINITY.EXE, that runs on a standard IBM-style personal computer (PC).

The Control Program is responsible for the management of all low-level hardware signals, fault detection, pulse energy monitoring and servo systems. It is important to realize that all direct access to the hardware is done by the Control Program stored in the EPROM. If the communications fail between the power supply and the PC, the laser remains in a safe operating state since the Control Program knows how to "take care of itself."

The User Interface program accepts input from the user in order to change the laser's state. It also displays diagnostic and status information about the laser. Its primary purpose is to interact with the user; it has no ability to exercise direct control over the laser hardware.

When the control program is upgraded it may be necessary to replace the EPROM in the power supply. Use the following procedure to do this.

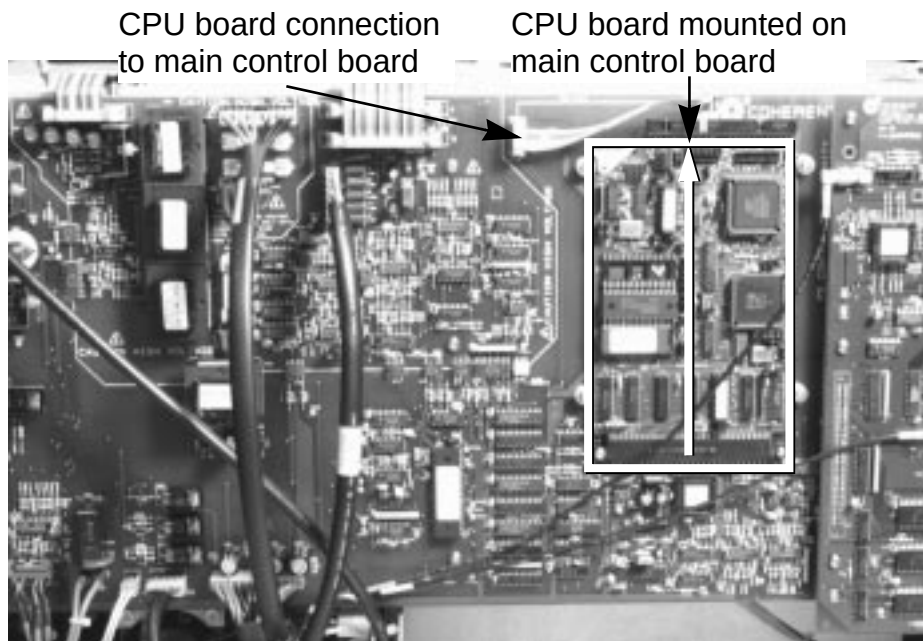
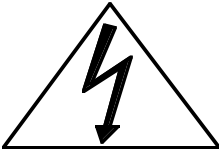


Figure 6-12. Side View of Control Board and CPU Board in Power Supply

1. Turn “off” the power supply, open the breaker on the back of the power supply, and switch off the main facility circuit breaker.



It is important to note that there is 3-phase power on the power supply control board even when the key switch is in the “off” position. Therefore, to minimize the risk of electrical shock, the fused breaker on the back of the Infinity power supply should be opened before the protective cover is removed.

2. Exit the Infinity program and turn off the computer.
3. Remove the smaller of the two power supply covers.
4. Locate the CPU board on the large control board, Figure 6-12. Disconnect the cable from the control board and slide the CPU board up and out of the power supply.
5. Carefully note how the old EPROM is mounted in its socket, Figure 6-13. The notch on the EPROM should match up with the drawing on the board and the EPROM must be positioned as far towards the left as possible in the socket.

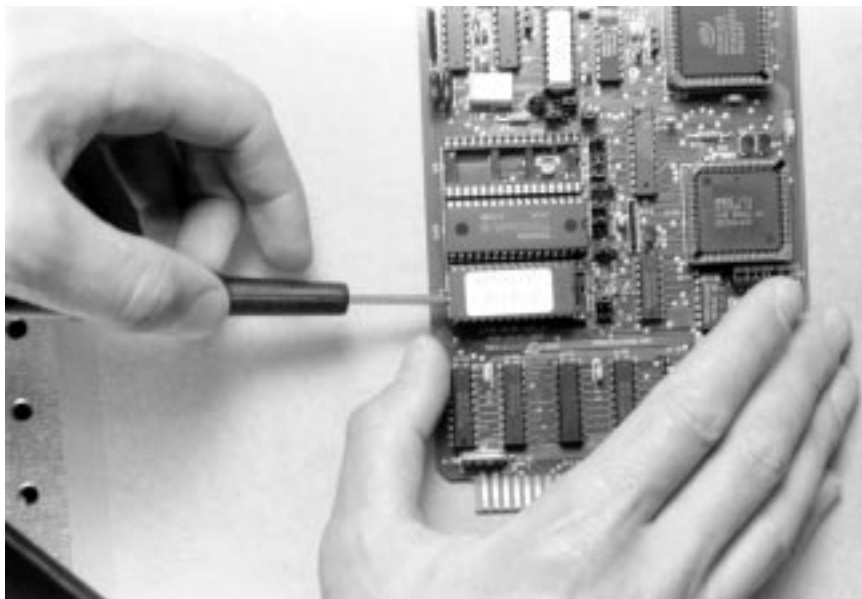


Figure 6-13. Removing the EPROM from the CPU Board

6. Remove the old EPROM using e.g. a small screwdriver as shown in Figure 6-13.

7. Insert the new EPROM in the socket the same way as the old chip was mounted.
8. Put the CPU board back on the control board. Connect the cable to the control board.
9. Put the power supply cover back on.
10. The EPROM replacement procedure is now completed and the laser is ready to be turned on again.

Lamp Timing

The Infinity lamp flash pulse width, and relative DPSS Q-switch pulse position, are set at the factory so that the laser performs optimally. Under normal usage of the laser there should be no need to change them.

The following procedure describes how to restore these parameters if they are lost. Note that these parameters can also be found on the software floppy disks and the final test “performance” sheet delivered with the laser system.



Modifying the parameters contained in the lamp timer window will affect laser performance. Deviation from factory settings may result in lower output power and unstable DPSS laser performance.

Due to internal delay in the large transistor that discharges the capacitor bank through the flash lamp, the set and the actual lamp pulse width differ, see Figure 6-14. For precise timing control an oscilloscope with one channel connected through inductive coupling to the “cathode” flash lamp lead and one to a photodiode monitoring the Q-switch pulse, can be used. This setup is illustrated in Figure 6-15.

It is desirable to set the Q-switch delay so that it opens just before the lamp is turned off. It has been found that positioning the Q-switch $\sim 4 \mu\text{s}$ before the falling edge of the lamp current is optimal.

Procedure

1. Open the Infinity head cover and install the interlock defeat. Tape a piece of electrical wire, $\sim 8 \text{ cm}$ long, in parallel to the white cathode flash lamp cable, Figure 6-15.
2. Click on the “Service...” button in the main window and then select “Lamp” from the Service window menu. In the resulting drop down menu select “Timing...”. Dismiss the warning window which opens when the Timing window is selected.
3. Connect the “Lamp Trig Synch” BNC output, on the front panel of the Infinity power supply, to the trigger channel of the oscilloscope. Set the scope trigger channel for 50Ω termination and to trigger off the falling edge of the power supply trigger signal.

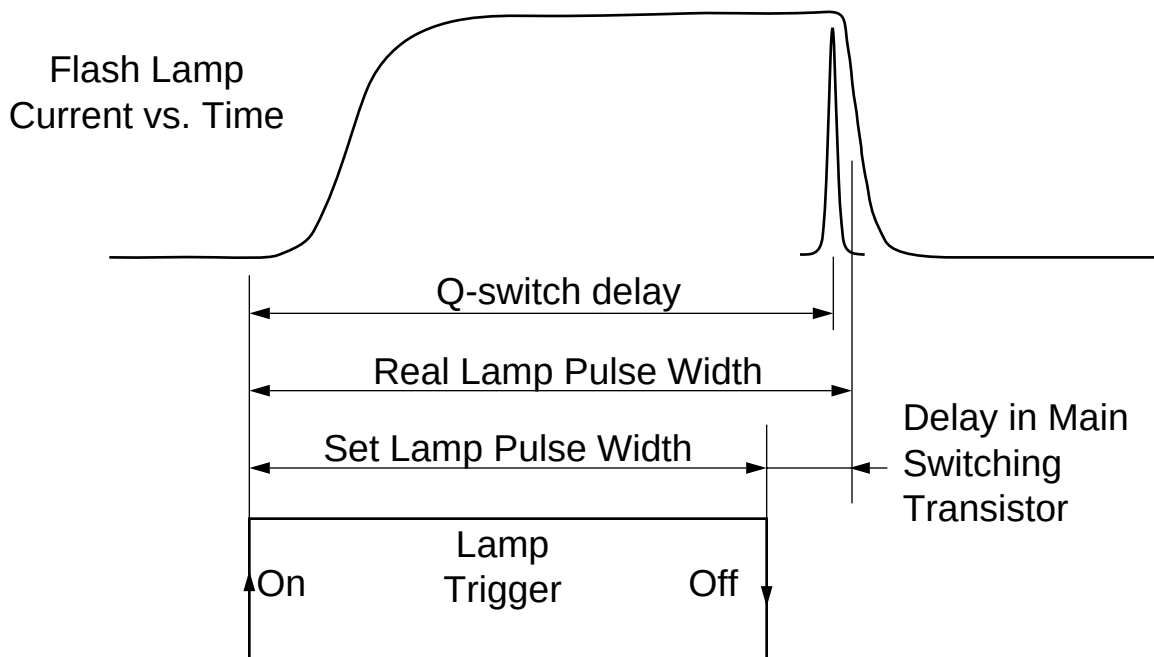


Figure 6-14. Timing of Lamp Trigger, Lamp Current and Q-switch.

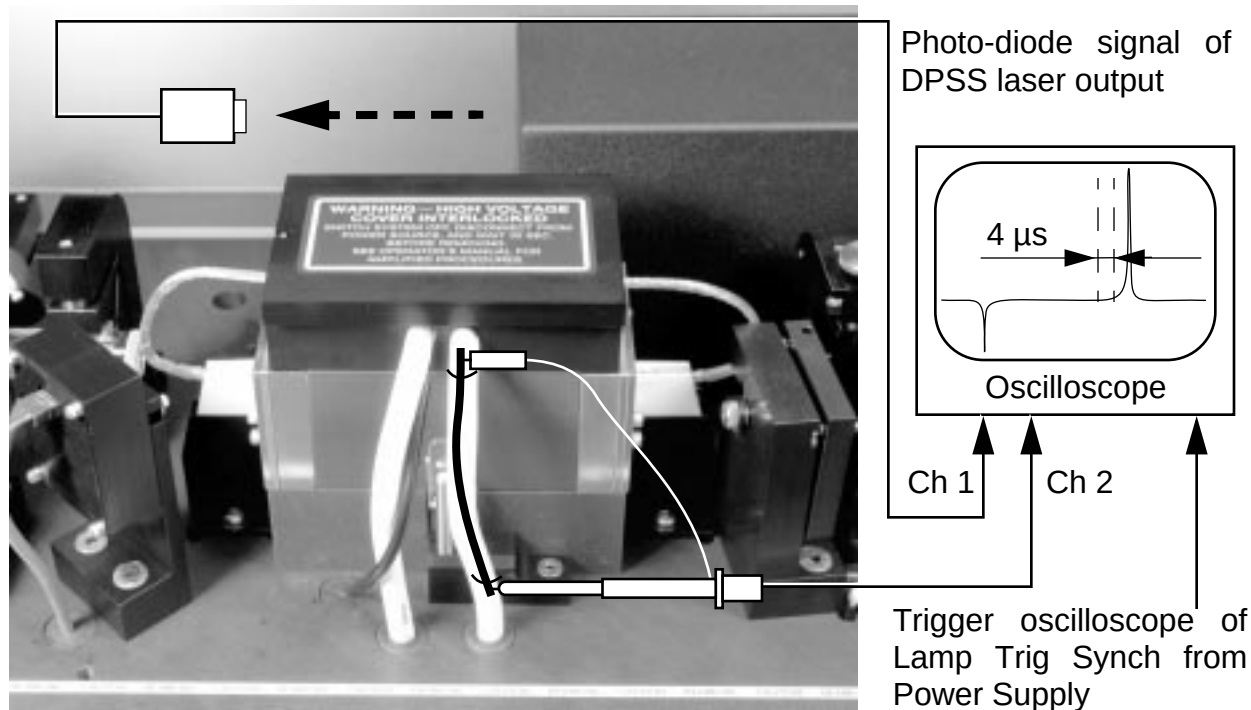


Figure 6-15. Setup to Determine Timing Settings for Lamp Pulse and Q-switch.

4. Connect an 1x oscilloscope probe across the wire attached to the lamp cathode lead. Terminate the probe at 1 M Ω . Place the IR viewing card (or a white business card) between the Faraday Isolator and the Collimating lens to block the DPSS output.
5. Set the system repetition rate to its maximum value and set the IR energy to 1 mJ. Start the Infinity laser system. The solid trace in Figure 6-15 shows the derivative of the lamp current (inductive coupling). This trace should be observed on the oscilloscope.
6. Using the “Lamp pulse width” slide bar of the timing window, set the lamp pulse to 250 μ s (the time between the lamp current turn on and turn off “spikes”).



To see the Q-switch pulse on the photodiode, it may be necessary to block the light from the flash lamp.

7. Hook the photodiode to the oscilloscope and slowly zoom in on the lamp turn off spike while looking for the Q-switch signal from the DPSS.
8. Using the Q-switch delay slide bar, position the laser pulse so that it occurs 4 μ s before the lamp current turns off.

After setting the system timing, closing the timing window will automatically save the new parameters.

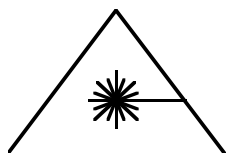
9. Turn off the Infinity laser system and remove the beam block from between the Faraday Isolator and the Collimating lens. Remove the 1x oscilloscope probe and the photodiode if it was placed inside the laser head. Remove the interlock defeat and close the Infinity head cover. The system should now be ready for use.

Safety and Diagnostic Equipment

The following is a list, including descriptions, of equipment which would prove to be helpful in the maintaining of the Infinity laser system.

Power Meters

An external power meter for the Infinity system should be able to handle an average power of 50 W and the high peak powers associated with pulsed Nd:YAG lasers. Be aware that a power meter can break even when exposed to intensities less than the specified damage threshold if not handled correctly. If suddenly exposed to a high power laser pulse it might break from the thermal shock. The recommended procedure is to position the meter in the beam at low power and then slowly turn up the power. If a power meter is available but does not have the necessary damage threshold, either a fraction of the beam can be split off using a glass slide or the beam can be expanded before entering the power meter. If the latter option is chosen it is advisable to use a negative focal length lens since this will avoid bringing the beam to a focus.



For some power meters the sensitive surface is highly reflective. Be aware of where the reflected beam is going, when positioning a power meter or an optical component in the beam path!

It is a good habit to keep a log book in which the laser power, lamp voltage and burn patterns of the fundamental and the harmonics, measured at regular intervals and after any changes, such as replacing the flash lamp, have been made to the system, are recorded. Power meters well suited to work with the Infinity system are sold by Coherent. Call 408-764-4983 or fax 408-764-4800 (within the U.S.) or your local Coherent representative for more information.

IR Viewing Cards

These cards provide an inexpensive means of determining the location of the invisible IR beam from the oscillator and a way to track it through the system. They are not suitable for the IR pulse since the power of this is enough to permanently burn the coating of the card. When using a IR card in the beam, keep it in constant motion to avoid burning it. As with all reflective surfaces put into the beam path, be aware of where the back reflex is going.

IR Viewing Scopes

These instruments usually look and are handled like a small camera. They produce an image in a view-finder that shows both the visible and the near infrared light present. Even though this device is not needed for the maintenance of the system, it is useful for making sure that no harmful IR beams are present in the room.

Beam Diagnostic Instruments

In order to characterize the beam and align the laser with high precision it is necessary to have an instrument showing the mode of the beam in detail. Such instruments usually consist of a small CCD camera, a frame grabber card and a computer. If the system is bought as a package it will also come with the software needed to read out the camera and to manipulate the image on the computer. A CCD camera system well suited to work with the Infinity system is sold by Coherent. Call 916-888-5107 or fax 916-885-6039 for more information.

Photo-Diodes

To monitor the temporal behavior of the laser a fast photo diode connected to an oscilloscope is sufficient. The photo diode should have a rise time of less than 1 ns and the oscilloscope a band width of 350 MHz or faster. Be aware that photo-diodes can be very sensitive and may easily be damaged by intense laser pulses. It is often sufficient to just position the photo diode in the vicinity of the laser where there is enough stray light to obtain a signal.

For advice on finding and selecting diagnostic and safety instruments for the Infinity laser system, please contact your local Coherent representative.

